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Lockeport's commercial fishing boats use gill nets, which kill many of the netted fish, including some fish of endangered species. The fishing commission has proposed requiring the use of tent nets, which do not kill fish; boat crews would then throw back fish of endangered species. Profitable commercial fishing boats in similar areas have already switched over to tent nets. The proposal can therefore be implemented without economic harm to Lockeport's commercial fishing boat operators.

Which of the following, if true, casts the most serious doubt on the argument made for the proposal?

A. In places where the use of tent nets has been mandated, there are typically fewer commercial fishing boats in operation than there were before tent nets came into use.

B. Even when used properly, gill nets require many more repairs than do tent nets.

C. Recreational anglers in Lockeport catch more fish of endangered species than do commercial fishing boats.

D. The endangered species of fish in Lockeport's commercial fishing area did not become endangered as a result of the use of gill nets by fishing fleets.

E. The endangered species of fish caught by Lockeport's commercial fishing fleet are of no commercial value.

Evaluation of a Plan

Situation Gill nets, used by Lockeport's commercial fishing boats, kill some fish of endangered species. The fishing commission has proposed requiring the use of tent nets, which do not kill fish. This would allow the fish of endangered species to be thrown back. It is argued that the proposed requirement will not harm commercial fishing boat operators, since commercial fishing boats in other similar places are using tent nets and are profitable.

Reasoning *What new piece of information would weaken the argument for the proposed requirement?* The crucial support given for the argument is that in other similar places, commercial fishing boats that use tent nets and not gill nets are profitable. But if, in places where only tent nets are now used, the numbers of commercial fishing boats diminished, it would be reasonable to suspect that switching entirely to tent nets may have driven some of the fishing operations out of business or caused them to move to other areas in which there was no expectation that they would use only tent nets.

A. **Correct.** This is new information. As explained above, it would justify doubt about the argument made in favor of the proposal.

B. This suggests that implementation of the proposed requirement could, over time, lower a certain type of operational costs for commercial fishing boats using tent nets. This would provide a new reason in support of the proposed requirement, not a reason to doubt the argument for it.

C. This suggests that perhaps recreational fishing in Lockeport needs to be regulated more strictly, but that is a separate issue from the one addressed in the argument for the proposed tent-net requirement.

D. This information casts no doubt on the relevance of the stated information that using gill nets contributes to undermining populations of at least some of the endangered fish species in Lockeport. If the species are currently endangered, they may need protection regardless of how they became endangered.

E. This information suggests that fish of the endangered species in Lockeport cannot profitably be sold. It does not cast doubt on the argument made in favor of the proposed requirement. Neither does it cast doubt on the practicality or the desirability of the proposed requirement.

The correct answer is A.

•

Normally, the pineal gland governs a person's sleep-wake cycle by secreting melatonin in response to the daily cycle of light and darkness as detected by the eye. Nonetheless, many people who are totally blind due to lesions in the visual cortex of the brain easily maintain a 24-hour sleep-wake cycle. So, the neural pathway by which the pineal gland receives information from the eye probably does not pass through the visual cortex.

For purposes of evaluating the argument, it would be most useful to establish which of the following?

A. Whether melatonin supplements help people who have difficulty maintaining a 24-hour sleep cycle to establish such a pattern.

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such a pattern

B. Whether the melatonin levels of most totally blind people who successfully maintain a 24-hour sleep-wake cycle change in response to changes in exposure to light and darkness

C. Whether melatonin is the only substance secreted by the pineal gland

D. Whether most people who do not have a 24-hour sleep-wake cycle nevertheless have a cycle of consistent duration

E. Whether there are any people with normal vision whose melatonin levels respond abnormally to periods of light and darkness

Argument Evaluation

Situation Normally, a person's sleep-wake cycle is governed by the pineal gland secreting melatonin in response to the daily cycle of light and darkness as detected by the eye. Yet many people who are totally blind due to lesions of the visual cortex easily maintain a 24-hour sleep-wake cycle.

Reasoning *What additional information would be most helpful in evaluating the argument?* The argument's conclusion is that the neural pathway by which the pineal gland receives information probably does not pass through the visual cortex. This is suggested by the fact that people without a well-functioning visual cortex (e.g., people with a certain type of blindness) can nonetheless maintain a 24-hour sleep-wake cycle. Is it by the pineal gland's secretion of melatonin that they do so? The argument tells us that *normally* (i.e., in sighted people), this is the mechanism for sleep regulation. But the argument depends on assuming that a similar mechanism is operating in people who are blind but have well-regulated sleep cycles. The best choice will be the one that helps us decide whether that assumption is correct.

A. This question would not give us an answer that would help in evaluating the argument. A "no" answer would not clarify whether the pineal gland–melatonin mechanism operates in people who are blind. A "yes" answer would do no better. The question refers only to people who have sleep dysfunctions (which the argument does not address).

B. **Correct.** Answering this question would provide the most useful information for evaluating the argument. A "yes" answer would help confirm a key assumption of the argument: that blind people rely on the pineal gland-melatonin mechanism for sleep regulation. A "no" answer would help disconfirm that assumption.

C. Whether or not there are other substances secreted by the pineal gland makes no difference to the reasoning. The argument relies on the premise that the pineal gland governs the sleep cycle *by secreting melatonin*. For example, if the pineal gland sometimes secreted adrenaline, that would still have no bearing on the argument.

D. The consistency or inconsistency of the duration of some people's sleep patterns has no relevance to the reasoning. Their sleep patterns could be due to any of a number of factors.

E. This does not help, for there could be sighted people whose melatonin levels respond abnormally simply because of a pineal-gland abnormality.

The correct answer is B.

•

In countries where automobile insurance includes compensation for whiplash injuries sustained in automobile accidents, reports of having suffered such injuries are twice as frequent as they are in countries where whiplash is not covered. Presently, no objective test for whiplash exists, so it is true that spurious reports of whiplash injuries cannot be readily identified. Nevertheless, these facts do not warrant the conclusion drawn by some commentators that in the countries with the higher rates of reported whiplash injuries, half of the reported cases are spurious. Clearly, **in countries where automobile insurance does not include compensation for whiplash, people often have little incentive to report whiplash injuries that they actually have suffered.**

In the argument given, the two **boldfaced** portions play which of the following roles?

A. The first is a claim that the argument disputes; the second is a conclusion that has been based on that claim.

B. The first is a claim that has been used to support a conclusion that the argument accepts; the second is that conclusion.

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C. The first is evidence that has been used to support a conclusion for which the argument provides further evidence; the second is the main conclusion of the argument.

D. The first is a finding whose implications are at issue in the argument; the second is a claim presented in order to argue against deriving certain implications from that finding.

E. The first is a finding whose accuracy is evaluated in the argument; the second is evidence presented to establish that the finding is accurate.

Argument Evaluation

Situation Reported whiplash injuries are twice as common in countries where car insurance companies pay compensation for such injuries as they are in countries where insurance companies do not. Although there is no objective test for whiplash, this does not mean, as some suggest, that half of the reports of such injuries are fake. It could simply be that where insurance will not pay for such injuries, people are less inclined to report them.

Reasoning *What roles do the two **boldfaced** portions play in the argument?* The first portion tells us about the correlation between reported cases of whiplash in countries and the willingness of insurance companies in those countries to compensate for whiplash injuries. The argument next states that whiplash is difficult to verify objectively. The argument then asserts that although this last fact, taken together with the first **boldfaced** portion, has led some to infer that over half of the reported cases in countries with the highest whiplash rates are spurious, such an inference is unwarranted. The second **boldfaced** portion then helps to explain, by offering an alternative explanation, why such an inference is not necessarily warranted.

A. The claim made in the first **boldfaced** portion is never disputed in the argument. The second is not the argument's conclusion.

B. Perhaps the argument uses the first portion to support its conclusion, but there is no indication that it has been used elsewhere to do so. Regardless, the second **boldfaced** portion is not the argument's conclusion.

C. The first portion has been used to support a conclusion that the argument rejects; the second portion is not the argument's conclusion.

D. **Correct.** This answer choice correctly identifies the roles played in the argument by the **boldfaced** portions. As explained in the Reasoning section above, the first **boldfaced** portion tells us about the correlation between reported cases of whiplash in countries and the willingness of insurance companies in those countries to compensate for whiplash injuries. This is presented as a fact whose implications are at issue in the ensuing portions of the passage. The argument then reports that this fact, considered together with the difficulty of proving whiplash injuries, has led some to infer that over half of the reported cases in countries with the highest whiplash rates are spurious. The conclusion of the argument is that this inference is unwarranted. The second **boldfaced** portion expresses the basis for the argument's conclusion: there is a reasonable alternative explanation for the differences in frequency of whiplash injury reports between the two types of countries.

E. The accuracy of the first **boldfaced** portion is never questioned in the argument, nor is the second intended to somehow help show that the first is accurate. Rather, the argument assumes that the first portion is accurate.

The correct answer is D.

•

Last year Comfort Airlines had twice as many delayed flights as the year before, but the number of complaints from passengers about delayed flights went up three times. It is unlikely that this disproportionate increase in complaints was rooted in an increase in overall dissatisfaction with the service Comfort Airlines provides, since the airline made a special effort to improve other aspects of its service last year.

Which of the following, if true, most helps to explain the disproportionate increase in customer complaints?

A. Comfort Airlines had more flights last year than the year before.

B. Last year a single period of unusually bad weather caused a large number of flights to be delayed.

C. Some of the improvements that Comfort Airlines made in its service were required by new government regulations.

D. The average length of a flight delay was greater last year than it was the year before.

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D. The average length of a flight delay was greater last year than it was the year before.

E. The average number of passengers per flight was no higher last year than the year before.

Argument Construction

Situation Last year Comfort Airlines had twice as many delayed flights as it did the year before but three times as many passenger complaints about delayed flights. The airline made a special effort to improve other aspects of its service last year.

Reasoning *What could explain why the number of complaints about delayed flights increased disproportionately to the number of delayed flights last year?* In other words, why did the average number of passenger complaints per delayed flight go up last year? One obvious possibility is that the average number of passengers per delayed flight was greater last year than it had been the year before. Another is that the flight delays tended to cause worse problems for passengers last year than they had the year before, so that on average each delay was more upsetting for the passengers.

A. This helps explain why the airline had more delayed flights last year, but not why the increase in complaints about delayed flights was disproportionate to the increase in delayed flights.

B. This helps explain why the airline had more delayed flights last year. But, if anything, the situation should have reduced the number of passenger complaints per delayed flight, since many passengers should have realized that the unusually bad weather was not the airline's fault.

C. If any of the improvements concerned handling of flight delays, for example, and passengers were aware that government regulations addressed this, then passengers might have complained more than previously. But the information we are given here is too general and too vague to explain the disproportionate increase in complaints.

D. **Correct.** Longer flight delays would have more severely inconvenienced passengers and thus would probably have generated more passenger complaints per delay.

E. This rules out the possibility that an increased number of passengers per delayed flight could have caused the disproportionate increase in the number of complaints about delayed flights. But no alternative explanation is offered.

The correct answer is D.

•

Last year a global disturbance of weather patterns disrupted harvests in many of the world's important agricultural areas. Worldwide production of soybeans, an important source of protein for people and livestock alike, was not adversely affected, however. Indeed, last year's soybean crop was actually slightly larger than average. Nevertheless, the weather phenomenon is probably responsible for a recent increase in the world price of soybeans.

Which of the following, if true, provides the strongest justification for the attribution of the increase in soybean prices to the weather phenomenon?

A. Last year's harvest of anchovies, which provide an important protein source for livestock, was disrupted by the effects of the weather phenomenon.

B. Most countries that produce soybeans for export had above-average harvests of a number of food crops other than soybeans last year.

C. The world price of soybeans also rose several years ago, immediately after an earlier occurrence of a similar global weather disturbance.

D. Heavy rains attributable to the weather phenomenon improved grazing pastures last year, allowing farmers in many parts of the world to reduce their dependence on supplemental feed.

E. Prior to last year, soybean prices had been falling for several years.

Argument Construction

Situation A weather disturbance last year disrupted harvests worldwide but did not reduce production of soybeans, a protein source for both people and livestock. Soybean prices increased nonetheless, likely a result of the weather.

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Reasoning *What evidence would suggest that the weather disturbance caused the increase in soybean prices even though it did not reduce soybean production?* Prices tend to increase when the supply of a product falls relative to the demand for the product. But the production of soybeans did not fall. Evidence that the weather disturbance either hindered the global distribution of soybeans or increased global demand for soybeans could support the claim that the weather disturbance caused the increase in soybean prices.

A. **Correct.** If the weather disturbance reduced the anchovy harvest, and anchovies provide protein for livestock just as soybeans do, then more soybeans for livestock feed would be needed to compensate for the lack of anchovies. The resulting increase in demand for soybeans could thus have increased global soybean prices.

B. This is not surprising, given that the weather disturbance did not severely affect the soybean-producing countries, but it does not explain how the weather disturbance could have caused soybean prices to increase.

C. The rise in soybean prices after the earlier weather disturbance could easily have been a coincidence. Or, unlike last year's disturbance, the earlier disturbance could have reduced soybean production.

D. This suggests that demand for soybeans should have fallen as a result of the weather disturbance, so it does not explain why soybean prices rose.

E. If soybean prices were unusually low for some temporary reason when the weather disturbance occurred, they might have been likely to rise back to normal levels even without the weather disturbance.

The correct answer is A.

•

Most of the year, the hermit thrush, a North American songbird, eats a diet consisting mainly of insects, but in autumn, as the thrushes migrate to their Central and South American wintering grounds, they feed almost exclusively on wild berries. Wild berries, however, are not as rich in calories as insects, yet thrushes need to consume plenty of calories in order to complete their migration. One possible explanation is that berries contain other nutrients that thrushes need for migration and that insects lack.

Which of the following, if true, most seriously calls into question the explanation given for the thrush's diet during migration?

A. Hermit thrushes, if undernourished, are unable to complete their autumn migration before the onset of winter.

B. Insect species contain certain nutrients that are not found in wild berries.

C. For songbirds, catching insects requires the expenditure of significantly more calories than eating wild berries does.

D. Along the hermit thrushes' migration routes, insects are abundant throughout the migration season.

E. There are some species of wild berries that hermit thrushes generally do not eat, even though these berry species are exceptionally rich in calories.

Argument Evaluation

Situation Hermit thrushes are songbirds that usually eat insects but switch to eating berries when migrating. The thrushes need lots of calories to migrate, but berries contain fewer calories than insects do. Perhaps the berries contain nutrients that insects do not provide.

Reasoning *What would cast doubt on the claim that the thrushes switch to berries because berries contain nutrients that insects lack and that the thrushes need for their migration?* Evidence that berries do not contain such nutrients or that thrushes do not decrease their net calorie consumption by eating berries would cast doubt on the proposed explanation. So would any evidence that supported an alternative explanation for the diet change during migration—for example, seasonal or regional differences in the amount or quality of berries or insects available for the thrushes to consume.

A. Even if thrushes need to be well-nourished to finish migrating before winter, extra nutrients found in berries but not insects might help provide the nourishment they need.

B. Even if insects contain *certain nutrients* not found in wild berries, those specific nutrients may not be the ones

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the thrushes need for their migration.

C. **Correct.** This suggests that the thrushes might gain more net calories from eating berries than from eating insects, which could explain why they switch to eating berries even if the berries contain no extra nutrients.

D. By ruling out a lack of insects to eat while migrating as an alternative explanation for why the thrushes switch to eating berries, this would support the proposed explanation.

E. The calorie-rich species of berries the thrushes do not eat might be poisonous or indigestible for them, even if the species of berries the thrushes do eat contain nutrients they need to migrate.

The correct answer is C.

•

The kinds of hand and wrist injuries that result from extended use of a computer while maintaining an incorrect posture are common among schoolchildren in Harnville. Computers are important to the school curriculum there, so instead of reducing the amount their students use computers, teachers plan to bring about a sharp reduction in the number of these injuries by carefully monitoring their students' posture when using computers in the classroom.

Which of the following would it be most useful to know in order to assess the likelihood that the teachers' plan will be successful?

A. Whether extended use of a computer while maintaining incorrect posture can cause injuries other than hand and wrist injuries

B. Whether hand and wrist injuries not caused by computer use are common among schoolchildren in Harnville

C. What proportion of schoolchildren in Harnville with hand and wrist injuries use computers extensively outside the classroom

D. Whether changes in the curriculum could reduce the schools' dependence on computers

E. What proportion of schoolchildren in Harnville already use correct posture while using a computer

Evaluation of a Plan

Situation Hand and wrist injuries from using computers while maintaining poor posture are common among schoolchildren in Harnville. Teachers plan to greatly reduce the number of such injuries by monitoring their students' posture while the students use computers in the classroom.

Reasoning *What would be most helpful to know to determine the likelihood that the teachers' plan will succeed?* The primary concern is the *posture* students adopt while using computers. To succeed, the teachers' plan must reduce the time students spend with poor posture while using computers and reduce it enough to greatly reduce the number of injuries. To know how likely this is, it would help to know how effectively the teachers will be able to monitor and improve their students' posture inside the classroom. But how many of the students use computers *outside of school* while maintaining poor posture, and how often do they do so? If many students do so quite often, they may develop hand and wrist injuries regardless of what happens in school.

A. The teachers do not plan to reduce any injuries other than hand and wrist injuries, so whether computer use with poor posture causes any such other injuries is irrelevant to the likelihood that their plan will produce its intended effect.

B. The plan being discussed concerns only the reduction of hand and wrist injuries caused specifically by computer use with poor posture, so the frequency of hand and wrist injuries from other causes is irrelevant to the likelihood that the plan will produce its intended effect.

C. **Correct.** If the students' school use of computers is a large part of their overall computer use, any retraining that accompanies the monitoring might have some effect on their posture and related injury rates overall. However, the greater the proportion of children with hand and wrist injuries who use computers extensively outside the classroom, the more children are likely to keep developing the injuries regardless of any monitoring at school, so the less effective the teachers' plan involving only computer use at school is likely to be.

D. Knowing whether this is the case might help in developing a potential alternative to the teachers' plan, but if it did, this would not help significantly toward assessing the likelihood that the actual plan will succeed. The teachers' actual plan involves monitoring computer use in school without reducing such use. Other possible

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teachers' actual plan involves monitoring computer use in school without reducing such use. Other possible means of achieving the plan's goal are not part of the plan and are therefore irrelevant to the likelihood that the teachers' actual plan will succeed.

E. The passage indicates that the proportion of the schoolchildren maintaining poor posture while using computers is high enough for many to develop hand and wrist injuries as a result. Whatever the exact proportion is, the teachers' plan may or may not succeed in reducing it.

The correct answer is C.

•

A certain cultivated herb is one of a group of closely related plants that thrive in soil with high concentrations of metals that are toxic to most other plants. Agronomists studying the growth of this herb have discovered that it produces large amounts of histidine, an amino acid that, in test-tube solutions, renders these metals chemically inert. Hence, the herb's high histidine production must be the key feature that allows it to grow in metal-rich soils.

In evaluating the argument, it would be most important to determine which of the following?

A. Whether the herb can thrive in soil that does not have high concentrations of the toxic metals

B. Whether others of the closely related group of plants also produce histidine in large quantities

C. Whether the herb's high level of histidine production is associated with an unusually low level of production of some other amino acid

D. Whether growing the herb in soil with high concentrations of the metals will, over time, reduce their concentrations in the soil

E. Whether the concentration of histidine in the growing herb declines as the plant approaches maturity

Argument Evaluation

Situation A certain herb and closely related species thrive in soil full of metals toxic to most plants. The herb produces much histidine, which makes those metals chemically inert. Histidine production, therefore, is largely what accounts for the herb's thriving in metal-rich soils.

Reasoning *What evidence would help determine whether the herb's histidine production is what enables it to thrive in metal-rich soils?* The argument is that since the herb's histidine chemically neutralizes the metals that are toxic to most plants, it must explain why the herb can thrive in metal-rich soils. To evaluate this argument, it would be helpful to know about the relationship between other closely related plant species' histidine production and the ability to thrive in metal-rich soils. It would also be helpful to know about any other factors that might plausibly explain why the herb can thrive in those soils.

A. Whether or not the herb thrives in metal-free soils, histidine production could enable it to thrive in soils that contain toxic metals.

B. **Correct.** If the closely related plants do not produce much histidine, whatever other factor allows them to thrive in metal-rich soils would likely account for why the herb thrives in those soils as well.

C. The given information suggests no particular reason to suppose that a low level of some unspecified amino acid would enable a plant to thrive in metal-rich soils.

D. The herb might absorb metals from any metal-rich soil it grows in, regardless of why it thrives in that soil.

E. Whether or not histidine concentrations in the herb decline as it approaches maturity, there could still be enough histidine in the growing herb to neutralize the metals and explain why it can grow in metal-rich soil.

The correct answer is B.

•

Many people suffer an allergic reaction to certain sulfites, including those that are commonly added to wine as preservatives. However, since there are several winemakers who add sulfites to none of the wines they produce, people who would like to drink wine but are allergic to sulfites can drink wines produced by these winemakers without risking an allergic reaction to sulfites.

Which of the following is an assumption on which the argument depends?

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A. These winemakers have been able to duplicate the preservative effect produced by adding sulfites by means that do not involve adding any potentially allergenic substances to their wine.

B. Not all forms of sulfite are equally likely to produce the allergic reaction.

C. Wine is the only beverage to which sulfites are commonly added.

D. Apart from sulfites, there are no substances commonly present in wine that give rise to an allergic reaction.

E. Sulfites are not naturally present in the wines produced by these winemakers in amounts large enough to produce an allergic reaction in someone who drinks these wines.

Argument Construction

Situation People who are allergic to certain sulfites can avoid risking an allergic reaction by drinking wine from one of the several producers that does not add sulfites.

Reasoning *On what assumption does the argument depend?* Drinking wine to which no sulfites have been *added* will not prevent exposure to sulfites if, for instance, sulfites occur naturally in wines. In particular, if the wines that do not have sulfites added have sulfites present naturally in quantities sufficient to produce an allergic reaction, drinking these wines will not result in an allergic reaction. The argument therefore depends on assuming that this is not the case.

A. The argument does not require this because the conclusion does not address allergic reactions to substances other than sulfites.

B. The argument specifically refers to "certain sulfites" producing allergic reactions. It is entirely compatible with certain other forms of sulfites not producing allergic reactions in anyone.

C. This is irrelevant. The argument does not claim that one can avoid having an allergic reaction to sulfites *from any source* just by restricting one's wine consumption to those varieties to which no sulfites have been added.

D. Once again, the argument's conclusion does not address allergic reactions to substances other than sulfites in wine.

E. **Correct.** As explained in the Reasoning section above, the argument relies on the assumption that sulfites are not naturally present, in quantities sufficient to cause allergic reactions, in the wines to which no sulfites are added. If this assumption is not made, then the fact that no sulfites are added to certain wines is not a good reason to believe that people with sulfite allergies who consume the wines will not have an allergic reaction; if there are enough sulfites that naturally occur in the wine, people who consume the wine may well have an allergic reaction despite the fact that no sulfites have been added.

The correct answer is E.

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